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100+ EXAM QUESTIONS WITH WELL-EXPLAINED ANSWERS.



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Google Cloud Professional cloud Architect 100+ Practice Questions with Answers & Explanations.

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The author of the E-book is a certified professional cloud architect, cloud security engineer and Cybersecurity analyst with years of experience in the field. you can reach him through this website or subscribe to his weekly newsletter where he shares nuggets and tips to grow as a cloud engineer.

Thank you for purchasing this Q/A e-book. this book is meant to help you with your revision while preparing for the Google cloud professional cloud architect certification exam. The questions in this E-book have been carefully chosen to mirror the exam format and topic weight across various exam-tested domains.

This practice exam should be a guide to determine your preparedness for the exam.

If you do well on these practice questions, target (Over 75%) you should feel confident to sit for and pass the Professional Cloud Architect Exam on your first attempt.

I wish you the very best in your upcoming certification exam.

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Q1. An online retailer is evaluating database options to store the analytics data from its experimental drone deliveries. You're currently using a small cluster of MongoDB NoSQL database servers. You want to move to a managed NoSQL database service with consistent low latency that can scale throughput seamlessly and can handle the petabytes of data you expect after expanding to additional markets. What should you do?

A. Extract the data from MongoDB, and insert the data into BigQuery.
B.Extract the data from MongoDB. Insert the data into Firestore using Datastore mode.
C. Create a Bigtable instance, extract the data from MongoDB, and insert the data into Bigtable.
D. Extract the data from MongoDB. Insert the data into Firestore using Native mode.

Q2. How can you enable interactive access to a Google Cloud instance?

A. Edit the metadata of the instance
B. Use SSH to connect to the instance
C. Change the instance type to "Interactive"
D. Change the IP allow list for the instance

Q3. Your application requires a service to process changes to a Firestore database. The service is triggered each time a document is added to a specific collection in the database. What is the best way to configure this?
A. Deploy the service as a Cloud Function, triggered by Firestore events.

B. Deploy the service on Cloud Run, and configure it to poll the Firestore database for changes.
C. Deploy the service as a Cloud Function, triggered by Pub/Sub messages.
D. Deploy the service on Cloud Run, and configure it to respond to HTTP requests.

Q4. What is the difference between a virtual machine and a virtual machine instance in Google Compute Engine?

A. A virtual machine is a running copy of the VM, while an instance describes what the emulated server should look like.
B. A virtual machine is a physical server, while an instance is a virtual copy of that server.
C. A virtual machine is an executable file, while an instance is a running copy of the executable.
D. A virtual machine is an object that describes the VM's configuration, while an instance is a running copy of that configuration.

Q5. A social media company's employees will use Google Workspace. Your current on-premises network cannot meet the requirements to connect to Google's public infrastructure. What should you do?

A. Connect the on-premises network to Google’s public infrastructure via a partner that supports Carrier Peering.
B. Order a Partner Interconnect from a Google Cloud partner, and ensure that proper routes are configured.
C. Connect the network to a Google point of presence, and enable Direct Peering.
D.Order a Dedicated Interconnect from a Google Cloud partner, and ensure that proper routes are configured.

Q6. You have several Compute Engine instances running NGINX and Apache for a web application. In your web server logs, many login failures come from a single IP address, which looks like a brute force attack. How can you block this traffic?

A. Ensure that an HTTP(S) load balancer is configured to send traffic to the backend Compute Engine instances running your web server. Create a Google Cloud Armor policy with a default rule action of "Allow." Add a new rule that specifies the IP address causing the login failures as the Condition, with an action of "Deny" and a deny status of "403," and accept the default priority (1000). Add the load balancer backend service's HTTP-backend as the target.
B. Ensure that an HTTP(S) load balancer is configured to send traffic to your backend Compute Engine instances running your web server. Create a Google Cloud Armor policy using the instance's local firewall with a default rule action of "Allow." Add a new local firewall rule that specifies the IP address causing the login failures as the Condition, with an action of "Deny" and a deny status of "403," and accept the default priority (1000).

C. Ensure that an HTTP(S) load balancer is configured to send traffic to the backend Compute Engine instances running your web server. Create a Google Cloud Armor policy with a default rule action of "Deny." Add a new rule that specifies the IP address causing the login failures as the Condition, with an action of "Deny" and a deny status of "403," and accept the default priority (1000). Add the load balancer backend service's HTTP-backend as the target.
D.Edit the Compute Engine instances running your web application, and enable Google Cloud Armor. Create a Google Cloud Armor policy with a default rule action of "Allow." Add a new rule that specifies the IP address causing the login failures as the Condition, with an action of "Deny" and a deny status of "403," and accept the default priority (1000).

Q7. An IoT company must run in a separate project from its APIs and web store. You want to use Identity and Access Management (IAM) to ensure a secure environment. How should you set up IAM?

A. Use separate service accounts for each component (social media app, APIs, and web store) with predefined or custom roles to grant access.
B.Use one service account for each component (social media app, APIs, and web store) with predefined or custom roles to grant access.
C.Use separate service accounts for each component (social media app, APIs, and web store) with basic roles to grant access.
D. Use one service account for each component (social media app, APIs, and web store) with basic roles to grant access.

Q8. Which data storage service provides data warehouse services for storing data but also offers an interactive SQL interface for querying the data?
A. Datalab
B. BigQuery
C. Dataproc
D. Cloud SQL

Q9. An offshore wind company has recently acquired another company operating in Italy to expand your market in Europe. You have to integrate their legacy systems into your company's systems. Your company uses GCP as the primary cloud to manage your applications.
Assuming your team is taking this opportunity to modernize parts of the legacy system, what is the best way to integrate this system into your company's systems?

A. Containerize the legacy applications using GKE and run them on the on-prem instances. Use Anthos for managing the applications
B. Create snapshot of the legacy systems and run them on compute instances on GCP
C. Containerize the legacy applications using GKE and run the GKE instances on GCP
D. Use transfer appliance to securely ship the data from on-prem systems to GCP

Q10. _____ is a managed microservices platform that allows you to just build your application into a container, then let Google worry about handling all the infrastructure management from there.
A. Service Mesh
B. Config Connector
C. Cloud Build
D. Cloud Run

Q11. A Financial institution is planning to migrate its on-premises internal software to cloud identity. Your team has already manually addressed several of the security and network risks outlined by your company's compliance rules. You now have to convince higher management that your application will be safer in cloud identity than it was on-premises. What controls would you put in place to ensure network security is taken care of and convince higher management to migrate?

A. Place required Firewall rules in the set up
B. Place Cloud Armor in the set up
C. No need to add security controls. Network Security is Google's responsibility.

D. Place a set of VPCs in the set up to gain a more secure environment.

Q12.Google Cloud's operations suite integrates several technologies, including monitoring, logging, error reporting, and debugging that are commonly implemented in other environments as separate solutions using separate products. What are key benefits of integration of these services?

A. Better for Google Cloud only so long as you don't need to monitor other applications or clouds
B. Reduces overhead, reduces noise, streamlines use, and fixes problems faster
C. Ability to replace one tool with another from a different vendor
D. Detailed control over the connections between the technologies

Q13. What is the purpose of the Cloud Trace service?
A. Reporting on application errors.
B. Reporting on Google Cloud resource consumption as part of managing performance.
C. Reporting on latency as part of managing performance.
D. Reporting on Google Cloud system errors.

c

Q14. You've re-architected a monolithic web application, so state is not stored in memory on the web servers, but in a database instead. This has caused slow performance when retrieving user sessions though. What might be the best way to fix this?

A. Move session state back onto the web servers and use sticky sessions in the load balancer.
B. Use a caching service like Memorystore for Redis.
C.Increase the number of CPUs in the database server.
D.Make sure all web servers are in the same zone as the database.

B

Q15. You're building a RESTful microservice. Which would be a valid data format for returning data to the client?

A.JSON
B.XML
C.All options are correct
D.HTML

C

Q16. What Google Cloud feature would be easiest to use to automate a build in response to code being checked into your source code repository?

A. Build triggers
B.Binary Authorization
C. CloudScheduler
D. Cloud run

A

Q17. A global ecommerce company has users all over the world. They need a database service that can provide low latency worldwide with strong consistency. Which service should they choose?

A. Firestore

B. Cloud SQL
C. Spanner
D. BigQuery

c

Q18. When you use Google Cloud Code to integrate Anthos with an IDE, _____ uses your local Docker installation to manage a single Kubernetes cluster on your development machine.
A. OpenShift
B. OpenStack
C. Minikube
D. Skaffold

c

Q19. An Engineering firm is using Firestore to store information about products, reviews, and user sessions. You'd like to speed up data access in a simple, cost-effective way. What would you recommend?
A.Move the data to Spanner.
B. Move the data to Cloud Bigtable.
C. Move the data to BigQuery.
D. Cache the data using Memorystore.

Q20. Coinbase global has developed an online crypto exchange service to serve customers globally. The exchange service has a n-tier architecture, such as Web, REST API and databases, and it is hosted in Google Cloud. You are the lead of a QA and testing team. To make sure the platform can still operate under heavy load and in any regional natural disaster scenario. How would you design your load and DR test strategy?

A. Deploy Locust onto GKE to run load test until the system fails and conduct DR test during regular business hours to evaluate possible business impact by configuring Cloud Load Balancer to cut off the HTTP traffic at the web tier
B. Deploy Locust onto GKE to run load test until the system fails and conduct DR test during off-business hours to reduce possible business impact by configuring Cloud Load Balancer to cut off the HTTP traffic at the Web tier and API layer
C. Deploy Locust onto GKE to run load test until the system stops taking new requests and conduct DR test during regular business hours to evaluate possible business impact by configuring Cloud Load Balancer to cut off the HTTP traffic at the Web tier and API layer
D. Deploy Locust onto GKE to run load test until the system stops taking new requests and conduct DR test during off-business hours to reduce possible business impact by configuring Cloud Load Balancer to cut off the HTTP traffic at the Web tier, API layer and TCP traffic at database layer

Q21. A Music distribution platform acquires customers by letting them download songs after submitting a simple online form. They are currently using GCP infrastructure to host their website and content and using cloud storage to store the reports.

They observed that customers in some locations are facing issues in downloading the content and they are losing potential customers. They also have observed few users accessing the reports without submitting the form and want to avoid such behavior.

What services in GCP should they be using, that will help them with the above-mentioned problems? (Choose two)

A. Enable multi-regional storage for cloud storage. This enables the reports to be available in multiple regions.

B. Enable multi-regional storage with turbo replication for cloud storage. This enables the reports to be available in multiple regions.
C. Use signed urls with limited expiry time to give access to the reports
D. Enable Cloud CDN for the cloud storage where the reports are stored

Q22. **A multinational bank runs their GCP infrastructure in the South Africa (africa-south1)region. They promise high service levels to their customers and cannot risk regional unavailability. The head of IT wants to have a disaster recovery plan for their Cloud SQL instances in case the region fails or becomes unavailable.**
Regulation requires that they keep the data in the South Africa region. What are the steps for a complete disaster recovery plan that will give them near zero Recovery Point Objective? (Choose three)

A. Create cross-region read replicas of the SQL instances in South Africa (africa-south1). If failover is required, make the cross-region read replica as primary.
B. Create a new standby instance in different zone in south Africa(africa-south1)
C. Create cross-region read replicas of the SQL instances in South Carolina (us-east1). If failover is required, make the cross-region read replica as primary.
D. Create a new cross-region replica in Doha (me-central1) and attach to the primary instance.

Q23. A Large web archive application with users all over the world and a lot of static content. Which load balancer configuration would likely be the most suitable?
A. TCP load balancer with SSL configured.
B. HTTP load balancer with SSL configured.
C. HTTP load balancer with SSL configured and the CDN enabled.
D. UDP load balancer with SSL configured and the CDN enabled.

Q24. **An ecommerce firm operates in multiple geographic locations. They operate websites under different domains in each geography with separate users and product catalog for each domain. They use cloud spanner instances for databases in regional and multi-region deployments. The scaling of Cloud Spanner is managed by the application teams in respective geographies. Except for marketing events and campaigns, they mostly operate at constant capacity with some buffer for any spikes in traffic. They think there is a variation in traffic between different geographic regions and they can save costs by scaling the cloud spanner instances accordingly.**
They asked you for a solution that will optimize their cloud spanner instances as per the traffic of the websites. What do you suggest?

A. Deploy a centralized Autoscaler tool for Cloud Spanner and set the configuration parameters for CPU utilization, storage per node and margins commonly for all Cloud Spanner Instances.
B. Create an alert in cloud monitoring when the CPU utilization or node storage are reaching 80% and create a function to automatically scale the Cloud Spanner instances.
C. Deploy a distributed Autoscaler tool for Cloud Spanner and let the application teams manage the configuration parameters to set limits for CPU utilization, storage per node and margins for their Cloud Spanner Instances.

D. Edit Cloud Spanner settings to enable auto scaling. Set the limits for CPU utilization and storage per node to scale Cloud Spanner as per the varying load.

Q25. A firm wants a secure, private connection between your network and a Google Cloud network. There is not a lot of volume, but the

connection needs to be extremely reliable. Which configuration below would you choose?
A.VPC peering
B.Cloud Interconnect
C. VPN
D. VPN with high availability and Cloud Router.

Q26. An advertising agency has been developing a new feature for a GCP-hosted web-based CRM product. The marketing team wants to know if the new feature can improve the user engagement as well as minimize the user impact if the new feature doesn't work as expected. The CRM product is designed and developed using the modern microservices architecture and it is running on GKE. How do you implement a solution to accomplish this goal in Google Cloud?

A. Use Spinnaker to deploy the new feature enabled CRM to a group of internal users to test first, use BigQuery to analyze the data collected through Cloud Monitoring and Google Analytics to decide if the new feature should be released to the external users or not
B. Use Spinnaker to deploy the new feature enabled CRM to a small group of external users to test first, use BigQuery to analyze the data collected through Cloud Monitoring and Google Analytics to decide if the new feature should be released to all the external users or not

C. Use Spinnaker to do a canary deployment of the CRM product with traffic split - 25% of the traffic is routed to the product with the new feature and 75% for the old product, then use BigQuery to analyze the data collected through Cloud Monitoring and Google Analytics to decide if the new feature should be released to all the users or not

D. Use Spinnaker to do a canary deployment of the CRM product with traffic split - 10% of the traffic is routed to the product with the new feature and 90% for the old product, then use BigQuery to analyze the data

collected through Cloud Monitoring and Google Analytics to decide if the new feature should be released to all the users or not

Q27. You are deploying an existing application that was written in .NET version 4.8. The application requires Windows servers, and you don't want to change it. Which should you use?

A.Compute Engine
B. GKE
C. App Engine
D.Cloud Functions

Q28. A photo sharing site has asked you to write a program that uses Vision API to check for inappropriate content in photos that are uploaded to a Cloud Storage bucket. Any photos that are inappropriate should be deleted. What might be the simplest, cheapest way to deploy in this program?

A. Compute Engine
B. App Engine
C. Cloud Functions
D. GKE

Q29. A Survey company provides geospatial data through APIs to their clients. The APIs are hosted via on-premise web servers. The company uses Fluentd to collect logs from the servers and other data sources.

The product managers in the company are looking for data to understand customer behavior and want to analyze the logs in real time. They are looking for a cloud-based solution to set up their analytics workload on the cloud.

The company has approached you for a solution that helps them to store, analyze and visualize log data. What do you recommend?

A. Collect the logs from Fluentd to cloud storage. Use Cloud Dataflow to process the logs and send the output to a BigQuery table. Create a dashboard in Data Studio to visualize the output for product managers.
B. Create a BigTable and populate it with output from Fluentd. Process data in BigTable and create a dashboard in Data Studio to visualize the output for product managers.
C. Use Cloud Dataflow to ingest logs from Fluentd output. Process the data and create reports in Google sheets that can be shared with the product managers.
D. Configure Fluentd output plugin to use BigQuery as a destination. Use BigQuery to analyze the data. Create a dashboard in Data Studio to visualize the output for product managers.

Q30. You have containerized multiple applications using Docker and have deployed them using Compute Engine VMs. You want to save on costs and simplify container management. What might you do?

A. Write Terraform scripts for all deployment.
B.Rewrite the applications to run in App Engine standard environment.
C.Rewrite the applications to run in Cloud Functions.
D. Migrate the containers to GKE

Q31. A global betting firm that uses Google Cloud Platform plans for its employees to work from home. It has requested a scalable, cost-effective solution that can enable encrypted desktop streaming so that employees can access corporate resources. Which of the following would you recommend?

A. Google workspace
B. Google Cloud Virtual Desktop
C. Create an encrypted connection to the office network
D. Enable Remote Desktop Protocol (RDP) to connect to remote desktops

Q32. You're creating a service and you want to protect it from being overloaded by too many client retries in the event of a partial outage. Which design pattern would you implement?
A. Overload feedback repudiation
B. Truncated exponential backoff
C. Lazy caching
D. Circuit breaker

Q33. Your company needs a relational database for a system that requires extremely high availability (99.999%). The system must run uninterrupted even in the event of a regional outage. Which database would you choose?
A. Firestore
B.Spanner
C. Cloud SQL
D. BigQuery

Q34. You are the architect leading a team building a 3-tier web application. You do not want the programmers to have access to production resources. What's the easiest way to do this in Google Cloud?

A.Create development and production projects, and don't give developers access to production.
B. Use different service accounts for production and development resources with your project.
C. Set up private access and Identity-Aware Proxy.
D.Create a firewall rule that blocks developer access to production servers and databases.

Q35. You have been assigned the role of cloud storage admin in your organization. What do you have to do to enable encryption when using Cloud Storage?

A.Enable encryption and upload a key.
B. Nothing as encryption is enabled by default.
C. Simply enable encryption when configuring a bucket.
D. Create an encryption key using Cloud Key Management Service and select it when creating a Cloud Storage bucket.

Q36. A fintech start-up provides portfolio management services for customers holding digital assets. To manage the digital assets, they take custody of their customer's private keys which are very sensitive and prone to hacking.
They plan to use the private keys of high value assets only when required and store them offline in their on-premise systems. They use

GCP as their primary infrastructure and would like to know how to access and update the private keys securely only when required. What is the best way to achieve this? (Choose three)

A. Create a VM in a separate VPC and use it to temporarily store the private keys. Create a peering connection with the main VPC to update the or access the private keys.
B. Store the private keys on premise and copy them to the VM through cloud VPN.
C. Copy the updated private keys back to on-prem system and delete the VM instance
D. Connect the on-prem system to the main VPC through cloud VPN.

Q37. You are building a serverless app and have made a minor fix to one of your services. You want to deploy the new version with no downtime. Which would you choose?
A. Rolling update
B. A/B test
C.Canary deployment
D.Blue/green deployment

Q38. A solar company deploying test environments using Compute Engine VMs. Some downtime is acceptable, and it is very important to deploy them as inexpensively as possible. What single thing below could save you the most money?
A.Preemptible machines
B. Sustained use discount
C. Committed use discount
D. Sole tenant nodes

Q39. You have deployed service on App Engine that has an availability SLO of 99%. What could you use to monitor whether you are meeting it?
A. Liveness probe
B. Readiness probe
C.Uptime check
D. Health check

Q40. You made a minor update to a microservice running on Cloud run and would like to test it in production by sending a small portion of requests to the new version. Which would you choose?
A. A/B testing
B. Canary deployment
C. Blue/green deployment
D. Rolling update

Q41. How does Google Cloud suggest that organizations use third-party identity providers to enable users access to Google Cloud with their corporate credentials?
A. By delegating responsibility to service accounts and groups
B. By implementing the principle of least privilege
C. By federating third-party identity providers with Google Cloud
D. By migrating unmanaged accounts to personal accounts

Q42. A development team in your organization has launched a vital application on a regional managed instance group that must have nearly zero downtime.

To ensure high availability, you now must configure a global load balancer to direct HTTP traffic across multiple zones. After configuring the backend then the frontend of the load balancer, you click Review and Finalize.
Assuming you have configured the load balancer correctly, what backend settings do you see upon review?

A. The Backend service is web-app-backend. The Endpoint protocol is HTTP. The health check is web-app-load-balancer-check. The Instance group is load-balancing-web-app-group.
B. The Backend service is web-app-backend. The Endpoint protocol is TCP. The health check is load-balancing-web-app-group. The Instance group is web-app-load-balancer-check.
C. The Backend service is web-app-backend. The Endpoint protocol is HTTP. The health check is load-balancing-web-app-group. The Instance group is web-app-load-balancer-check.
D. The Backend service is web-app-backend. The Endpoint protocol is TCP. The health check is web-app-load-balancer-check. The Instance group is load-balancing-web-app-group.

Q43. As the principal cloud architect in your company, you want to compare resource utilization for VMs used for production, development, and testing. What should you do?

A. Export all machine logs to Cloud Storage and use Cloud Functions to build reports based on the VM tags.
B. Name the VMs with a prefix like “dev-”, “test-”, and “prod-” and filter on the name property when reporting.
C. Add a label called “state” to your VMs with the values “dev”, “test”, and “prod” and group by that label in your monitoring chart.
D.Put those resources in different projects and use dataflow to create an aggregation of log values for each.

Q44. You want to provide an external auditor with access to your logs, what IAM role would be best?
A. Logging Admin
B. Project Editor
C. Project Viewer
D. Logging Viewer

Q45. There is one application that has been utilizing a lot of bandwidth - sending out large packets. This app attempts to control the TCP window size so that it can maximize its own performance, to the detriment of other services running on the same WM.
Which Linux tunable below would you adjust to set the maximum OS send buffer size for all connections?

A. Net.core.rmem_max
B. Net.core.wmem_max
C. net.ipv4.tcp_rmem
D. net.ipv4.tcp_wmem

Q46. **Your company has migrated data to the Google Cloud Platform, and you are ready to connect your on-premise networks to Google Cloud. Review the requirements below and choose the best connection option.**

- **Guaranteed Network Availability and High Bandwidth: low**
- **Private Network connection: Not required**
- **Custom Routing: Not required**
- **Google Workspace Access: Required**
- **Budget to establish connection: None**

Which option fits the requirements best?

A. Direct Peering
B. Carrier Peering
C. Dedicated Interconnect
D. Partner Interconnect

Q47. You are the lead principal architect in charge of a firm that is migrating its on-premises data center to Google Compute Engine and need to calculate the number of vCPUs to support their current Intel Xeon CPU processing workload. Your data center has four servers, each with two dual-core CPUs. How many vCPUs would your instances need in total to support the same workload without compromising performance?
A.8
B.16
C.32
D.64

Q48. A medium-sized organization needs to use a tool to deploy its infrastructure. They want something that allows for repeatable deployment processes, uses a declarative language, and allows parallel deployment. They also want to deploy infrastructure as code on Google Cloud and other cloud providers. What should they do?
A. Automate the deployment with Cloud Deployment Manager.
B.Develop in Docker containers for portability and ease of deployment.
C. Automate the deployment with Terraform scripts
D. Use Google Kubernetes Engine (GKE) to create deployments and manifests for your applications.

Q49. You have been tasked by a large betting firm to deploy a load balancer for a web-based application with multiple backends in different regions. You want to direct traffic to the backend closest to the

end user, but also to different backends based on the URL the user is accessing. Which of the following could be used to implement this?

A. The request is received by the global external HTTP(S) load balancer. A global forwarding rule sends the request to a target proxy, which checks the URL map and selects the backend service. The backend service sends the request to Compute Engine instance groups in multiple regions.
B. The request is received by the SSL proxy load balancer, which uses a global forwarding rule to check the URL map, then sends the request to a backend service. The request is processed by Compute Engine instance groups in multiple regions.
C. The request is matched by a URL map and then sent to a SSL proxy load balancer. A global forwarding rule sends the request to a target proxy, which selects a backend service and sends the request to Compute Engine instance groups in multiple regions.
D.The request is matched by a URL map and then sent to a global external HTTP(S) load balancer. A global forwarding rule sends the request to a target proxy, which selects a backend service. The backend service sends the request to Compute Engine instance groups in multiple regions.

Q50. You are working for a chain of supermarkets where multiple GCP projects are used by several teams. While working on one of the projects, you faced some difficulties deploying a compute engine using the Deployment Manager. Your colleagues are no help, and you have contacted Google Support for the aid, but their representative is asking for access to project resources. How would you do that?

A. Add the email of the support executive in the IAM Users and provide the necessary roles.
B. Go to security tab followed by Access Approvals then Click on Approve after reviewing the details
C. Google Support has already access to all the resources of your project.
D. There is no way to provide access to the support team.

Q51 An Education platform wants to create a pipeline to automate the building of new application releases. What sequence of steps should you use?

A. Set up a source code repository. Run unit tests. Check in code. Deploy. Build a Docker container.
B. Set up a source code repository. Check in code. Run unit tests. Build a Docker container. Deploy.
C.Run unit tests. Deploy. Build a Docker container. Check in code. Set up a source code repository.
D. Check in code. Set up a source code repository. Run unit tests. Deploy. Build a Docker container.

Q52. Your organization wants a layered approach to security when setting up Compute Engine instances. What are some options you could use to make your Compute Engine instances more secure?

A. Use labels to allow traffic only from certain sources and ports. Use a Compute Engine service account.
B. Use network tags to allow traffic only from certain sources and ports. Use a Compute Engine service account.
C. Use labels to allow traffic only from certain sources and ports. Turn on Secure boot and vTPM.
D. Use network tags to allow traffic only from certain sources and ports. Turn on Secure boot and vTPM.

Q53. A security firm wants to allow partners to make orders programmatically, without having to speak on the phone with an agent. What should you consider when designing the API?

A. The API backend should be loosely coupled. Clients should not be required to know too many details of the services they use. For REST APIs, HTTP(S) is the most common protocol.

B. The API backend should be loosely coupled. Clients should not be required to know too many details of the services they use. REST APIs using gRPC should be used for all external APIs.
C. The API backend should be tightly coupled. Clients should know a significant amount about the services they use. For REST APIs, HTTP(S) is the most common protocol used.
D. The API backend should be tightly coupled. Clients should know a significant amount about the services they use. REST APIs using gRPC should be used for all external APIs.

Q54. An international Bank must meet compliance requirements. You as the architect need to ensure that employees with valid accounts cannot access their VPC network from locations outside of its secure corporate network, including from home. You also want a high degree of visibility into network traffic for auditing and forensics purposes. What should you do?

A. Enable Identity-Aware Proxy (IAP) to allow users to access services securely. Use Google Cloud's operations suite to view audit logs for the networks you need to monitor.
B. Enable VPC Service Controls, and use Google Cloud's operations suite to view audit logs for the networks you need to monitor.
C. Ensure that all users install Cloud VPN. Enable VPC Flow Logs for the networks you need to monitor.
D. Enable VPC Service Controls, define a network perimeter to restrict access to authorized networks, and enable VPC Flow Logs for the networks you need to monitor.

Q55. Your existing application runs on Fedora Linux VMs in an on-premises hypervisor. You want to deploy the application to Google Cloud with minimal refactoring. What should you do
A. Isolate the core features that the application provides. Use App Engine to deploy each feature independently as a microservice.

B. Write Terraform scripts to deploy the application as Compute Engine instances.
C. Set up a Google Kubernetes Engine (GKE) cluster, and then create a deployment with an autoscaler.
D. Use a Dedicated or Partner Interconnect to connect the on-premises network where your application is running to your VPC: Configure an endpoint for a global external HTTP(S) load balancer that connects to the existing VMs.

Q56. The dev team managing a no of compute engine VMs has submitted a ticket for a web application that is crashing due to high CPU utilization. The network admin has suggested creating a Managed Instance Group to handle the load. Which suggested configuration below could solve the problem?
A. Create a Managed Instance Group with an autoscaling policy based on load balancing serving capacity.
B. Create a Managed Instance Group with a cluster autoscaler with a fixed minimum and a maximum number of instances.
C. Create a Managed Instance Group with an autohealing policy that attempts to recreate the crashed instance.
D. Create a Managed Instance Group with an autoscaling policy based on CPU utilization.

Q57. A microservice has deployed its frontend web application in Kubernetes. Based on historical use, you need three pods to handle normal demand. Occasionally your load will roughly double. A load balancer is already in place. How could you configure your environment to efficiently meet that demand?
A. Edit your pod's configuration file and change the number of replicas to six.
B. Edit your deployment's configuration file and change the number of replicas to six.
C. Use the "kubectl autoscale" command to change the pod's maximum number of instances to six.

D. Use the "kubectl autoscale" command to change the deployment's maximum number of instances to six.

Q58.A movie renting business user account management app allows users to delete their accounts whenever they like. The movie business also has a very generous 60-day return policy for users. The customer service team wants to make sure that they can still refund or replace items for a customer even if the customer's account has been deleted. What can you do to ensure that the customer service team has access to relevant account information?

A. Ensure that the user clearly understands that after they delete their account, all their information will also be deleted. Remind them to download a copy of their order history and account information before deleting their account. Have the support agent copy any open or recent orders to a shared spreadsheet.
B. Disable the account. Export account information to Cloud Storage. Have the customer service team permanently delete the data after 30 days.
C. Temporarily disable the account for 30 days. Export account information to Cloud Storage, and enable lifecycle management to delete the data in 60 days.
D. Restore a previous copy of the user information database from a snapshot. Have a database administrator capture needed information about the customer.

Q59. An ecommerce firm is managing a supply chain product globally accessible. The firm has several partners across different countries that need to access BigQuery country-specific tables for historic data analysis for each country.
What should you do to guarantee that each partner accesses only their specific tables on BigQuery without altering the data?

A. Create a single dataset for all the countries, grant the role roles/bigquery.metadataViewer at table table-level for each country-specific tables for each partner
B. Create a single dataset for all the countries, grant the role roles/bigquery.dataViewer at table table-level for each country specific tables for each partner
C. Create a single dataset for all the countries, grant the role roles/bigquery.dataOwner at table table-level for each country specific tables for each partner
D. Create a single dataset for all the countries, grant the role roles/bigquery.dataEditor at table table-level for each country specific tables for each partner

Q60. Your Firm has hired an analytics firm that specializes in optimizing Google cloud costs. The analytics firm needs the billing information to analyze the cost patterns and come up with recommendations to save costs.
The head of IT is concerned about sharing the billing permissions to the analytics firm and asked for your advice. What would you recommend?

A. Create a Billing Account Viewer role at organizational level and assign it to the analytics firm. They should have access to the required billing information
B. Export cloud billing data to a BigQuery dataset. Create a BigQuery Data Viewer for the data set and assign it to the analytics firm. They should have access to the required billing information
C. Create a Billing Account User role at organizational level and assign it to the analytics firm. They should have access to the required billing information
D. Create a Billing Account Administrator role at organizational level and assign it to the analytics firm. They should have access to the required billing information

Q61. Your organization is experiencing success using Google Cloud and you want to leverage tools to make your solutions more efficient. Susan, one of the original web developers, currently adds new products to your application manually. Susan has many responsibilities and requires a long lead time to add new products. You need to create an App Engine application to let Your organizations employees add new products instead of waiting for Susan. However, you want to make sure that only authorized employees can use the application. What should you do?

A. Create a Google group and add authorized employees to it. Configure Identity-Aware Proxy (IAP) to the App Engine application as a HTTP-resource. Add the group as a principle with the role "Project Owner."

B. Create a Google group and add authorized employees to it. Configure Identity-Aware Proxy (IAP) to the App Engine application as a HTTP-resource. Add the group as a principle with the role "IAP-secured Web App User."
C.Use Google Cloud Armor to restrict access to the corporate network's external IP address. Configure firewall rules to allow only HTTP(S) access.

D. Set up Cloud VPN between the corporate network and the Google Cloud project's VPC network. Allow users to connect to the App Engine instance.

Q62. Your client created an Identity and Access Management (IAM) resource hierarchy with Google Cloud when the company was a startup. Your client has grown and now has multiple departments and teams. You want to recommend a resource hierarchy that follows Google-recommended practices. What should you do?
A.Keep all resources in one project, but change the resource hierarchy to reflect company organization.
B. Keep all resources in one project, and use a flat resource hierarchy to reduce complexity and simplify management.
C. Use multiple projects with established trust boundaries, and change the resource hierarchy to reflect company organization.

D.Use a flat resource hierarchy and multiple projects with established trust boundaries.

Q63. An online dating site is planning to launch much anticipated features to their existing application. They are using BigTable as a database and an application layer running on GKE containers. Since it is hard to predict the traffic, they are planning to use Ingress for load balancing to dynamically scale the application layer.
They want to restrict the traffic from the web layer to be restricted to the application layer and not the database layer.
What is the easiest way to achieve this?

A. Create a different subnet to each layer. Use VPC network peering to allow traffic between web layer & application layer and application & database layer
B. Create custom tags to identify the GKE containers and create firewall rules using the tags to allow the desired traffic flow
C. Create custom tags to identify the GKE containers and set up routes using the tags to allow the desired traffic flow
D. Create custom labels to identify the GKE containers and create firewall rules using the labels to allow the desired traffic flow

Q64. Your organization is legally required to comply with the Payment Card Industry Data Security Standard (PCI-DSS). Your organization has formal audits already, but the audits are only done periodically. The client needs to monitor for common violations to meet those requirements more easily. Your organization does not want to replace audits but wants to engage in continuous compliance and catch violations early. What would you recommend that this client do?

A. Enable the Security Command Center (SCC) dashboard, asset discovery, and Security Health Analytics in the Standard tier. Export or view the PCI-DSS Report from the SCC dashboard's Compliance tab.

B. Enable the Security Command Center (SCC) dashboard, asset discovery, and Security Health Analytics in the Premium tier. Export or view the PCI-DSS Report from the SCC dashboard's Vulnerabilities tab.
C. Enable the Security Command Center (SCC) dashboard, asset discovery, and Security Health Analytics in the Premium tier. Export or view the PCI-DSS Report from the SCC dashboard's Compliance tab.
D. Enable the Security Command Center (SCC) dashboard, asset discovery, and Security Health Analytics in the Standard tier. Export or view the PCI-DSS Report from the SCC dashboard's Vulnerabilities tab

Q65. A global social media site needs to make sure its new social media integration service can't be accessed directly from the public internet. They want to allow access only through the web front-end store. How can they prevent access to the social media integration service from the outside world, but still allow access to the APIs of social media services?

A. Limit access to the external IP addresses of the VM instances using firewall rules and place them in a private VPC behind Cloud NAT. Any SSH connection for management should be done with Identity-Aware Proxy (IAP) or a bastion host (jump box) after allowing SSH access from IAP or a corporate network.
B. Limit access to the external IP addresses of the VM instances using a firewall rule to block all outbound traffic. Any SSH connection for management should be done with Identity-Aware Proxy (IAP) or a bastion host (jump box) after allowing SSH access from IAP or a corporate network.
C. Remove external IP addresses from the VM instances running the social media service and place them in a private VPC behind Cloud NAT. Any SSH connection for management should be restricted to corporate network IP addresses by Google Cloud Armor.
D. Remove external IP addresses from the VM instances running the social media service and place them in a private VPC behind Cloud NAT. Any SSH connection for management should be done with Identity-Aware Proxy (IAP) or a bastion host (jump box) after allowing SSH access from IAP or a corporate network.

Q66. You have an application implemented on Compute Engine in the US-east1 region. You want to increase the durability of your application. What should you do?

A. Perform health checks on your Compute Engine instances.
B.Implement a regional managed instance group.
C. Implement a scheduled snapshot on your Compute Engine instances.
D. Monitor your application's usage metrics and implement autoscaling.

Q67. XJ drone company has implemented a manual CI/CD process for the container services required for the next implementation of XJ Drone Delivery project. You want to automate the process. What should you do?
A. Implement and reference a source repository in your Cloud Build configuration file.
B.Specify the name of your Container Registry in your Cloud Build configuration.
C. Configure and push a manifest file into an environment repository in Cloud Source Repositories.
D. Implement a build trigger that applies your build configuration when a new software update is committed to Cloud Source Repositories.

Q68. You have an application running on Compute Engine. You want to increase the durability of your application. What should you do?
A. Perform health checks on your Compute Engine instances.
B. Implement a regional managed instance group.
C. Implement a scheduled snapshot on your Compute Engine instances.
D.Monitor your application's usage metrics and implement autoscaling.

Q69. You are the lead architect in a team with developers who frequently write new versions of the code for one of your applications. You want to automate the build process when updates are pushed to Cloud Source Repositories. What should you do?
A. Set proper permissions for Cloud Build to access deployment resources.
B. Implement a build trigger that references your repository and branch
C. Implement a Cloud Build configuration file with build steps.
D. Upload application updates and Cloud Build configuration files to Cloud Source Repositories.

Q70. The number of requests received by your application is nearing the maximum specified in your design. You want to limit the number of incoming requests until the system can handle the workload. What design pattern does this situation describe?
A. Applying exponential backoff
B. Applying graceful degradation
C. Applying a circuit breaker
D.Increasing jitter

Q71. An airline wants to improve its pilot interface. They want to collect feedback on proposed changes from the community of pilots before rolling out updates systemwide. What type of deployment pattern should they implement?
A. You should implement a blue/green deployment.
B. You should implement A/B testing.
C. You should implement canary testing.
D.You should implement an in-place release.

Q72. A major clothes retailer has developed a data pipeline to ingest sales data from both physical stores and their website. The data scientist team recently trained a sales forecast model and deployed it into production. Over time, the model's performance has degraded. As

an architect, you need to help design a solution in Google Cloud to address the model performance issue. How would you implement a solution in GCP? (Choose Two)
A. Deploy the model to AI platform and set up a continuous evaluation job to detect model performance issue, then retrain the model
B. Deploy the model to AI platform and set up a model monitoring job to alert performance issue, then retrain the model
C. Deploy the model to Vertex AI and set up a continuous evaluation job to alert model performance issue, then retrain the model
D. Deploy the model to Vertex AI and set up a model monitoring job to alert data skew or data drift issue, then retrain the model

Q73. _____ allows you to use a simple YAML configuration file to define how all your containers interact with each other and with external network connections.
A. Config Management
B. Anthos Service Mesh
C. Cloud Run
D. Anthos on bare metal

Q74. An iron ore mining company has more than 1 million sensors installed in their vehicles all over the world. They collect some critical data that is transmitted in real time for fleet management. The rest of the data is uploaded daily when the vehicles reach their home base. The CIO is planning to launch an app that gives insights about vehicle health and usage based on the daily collected metrics for machine operators.
How do you design the app so that there is minimum latency for the functionality? (Choose two)

A. Upload the data from the vehicles to GCP and store it in cloud storage. Use dataflow connector to export required variables into a BigTable
B. Execute queries against the BigTable indexed by vehicle id and timestamp to display the insights on the app.

C. Upload the data from the vehicles to GCP and store it in cloud storage. Upload the data to Bigquery.
D. Execute queries against BigQuery indexed by vehicle id and timestamp to display the insights on the app.

Q75. **A university hosts many applications on GCP that each store various kinds of data. The head of data compliance is concerned that if there is any PII (personally identifiable information) present in the data, it might attract fines if not processed correctly.**
He asked you to come up with a solution that identifies the PII, classifies it and anonymizes it if it must be processed. He is aware of the dynamic nature of the applications and data and is looking for a solution that is easy to manage with minimum overhead.
As a GCP cloud architect what solution will you propose? (Choose two)

A. Use Cloud DLP (Data Loss Prevention) to scan data stored in Cloud Storage, Datastore and BigQuery & use Cloud Data Catalog to tag PII data
B. Use Cloud Data Catalog to search the PII in Cloud Storage, Datastore and BigQuery
C. Use Cloud DLP de-identification transformation templates for the datasets that are identified with PII & Cloud Dataflow to de-identify them
D. Use Cloud DLP to de-identify the PII datasets and use them for processing

Q76. A marketing agency is considering using Microsoft Active Directory in Google Cloud to centrally manage their users and applications. It is important that the server is highly available and fault tolerant as the employees use it globally to connect to different applications.
As a GCP cloud architect what solution will you recommend to the firm?

A. Create a custom mode VPC network with 2 subnets spanning 2 zones. Create Windows Server virtual instances and enable Active Directory Domain Services. Configure a new domain with Active Directory. Join the

new Windows Server instances to the new domain and configure firewall rules to allow traffic to the virtual machines.
B. Create a VPC with one subnet spanning 2 zones. Create Windows Server virtual instances and enable Active Directory Domain Services. Configure a new domain with Active Directory. Join the new Windows Server instances to the new domain and configure firewall rules to allow traffic to the virtual machines.
C. Create a custom mode VPC network with 2 subnets spanning 2 zones. Create Windows server virtual instances and enable Active Directory Domain Services. Create a load balancer with auto scaling group policy to run two virtual machines.
D. Create a VPC with one subnet spanning 2 zones. Create Windows server virtual instances and enable Active Directory Domain Services. Create a load balancer with auto scaling group policy to run two virtual machines.

Q77. A file backup company has adopted a multi-cloud strategy that uses a virtual machine-based infrastructure. The backup company website serves users across the globe. The company needs a single dashboard view to monitor performance in their AWS and Google Cloud environments. Your company previously experienced an extended outage and wants to establish a monthly service level objective (SLO) of no outage longer than an hour. What should they do?

A. Authorize access to your Google Cloud project from AWS with a service account. Install the monitoring agent on AWS EC2 (virtual machines) and Compute Engine instances. Use Cloud Monitoring to create dashboards that use the performance metrics from virtual machines to ensure that the SLO is met.
B. In Cloud Monitoring, create an uptime check for the URL your clients will access. Configure it to check from multiple regions. Use the Cloud Monitoring dashboard to view the uptime metrics over time and ensure that the SLO is met. Recommend an SLO of 97% uptime per month.
C. In Cloud Monitoring, create an uptime check for the URL your clients will access. Configure it to check from multiple regions. Use the Cloud

Monitoring dashboard to view the uptime metrics over time and ensure that the SLO is met. Recommend an SLO of 97% uptime per day.
D. Create a new project to use as an AWS connector project. Authorize access to the project from AWS with a service account. Install the monitoring agent on AWS EC2 (virtual machines) and Compute Engine instances. Use Cloud Monitoring to create dashboards that use the performance metrics from virtual machines to ensure that the SLO is met.

Q78. A Car rental service is using google cloud infrastructure for their compute and storage requirements. To comply with regulations, the company should store all the files available for audit by the authorities. All the files up to past 3 years should be available for audit before deletion.
The employees are aware of the regulatory requirements, but sometimes they might delete the files accidentally. The company wants to avoid such accidents as they attract fines from the authorities.
How should they manage the storage to be compliant with the regulation?

A. Use cloud storage for storing data. Enable multi region buckets and back-up data to a nearline storage. Set object lifecycle policy to delete objects that are more than 3 years old.
B. Use cloud storage for storing data. Add retention policy to the buckets with retention period of 3 years.
C. Use cloud storage for storing data. Enable multi region buckets and back-up data to a coldline storage. Set object lifecycle policy to delete objects that are more than 3 years old.
D. Use cloud storage for storing data. Enable object versioning for the buckets. Use object lifecycle policy to delete objects that are more than 3 years old.

Q79. A software company releases new patches to its software every 1.5 to 2 months. Although most releases are successful, you have experienced three problematic releases that caused major features to

be unavailable while software developers rolled back the release. You want to increase the reliability of software releases and prevent similar problems in the future. What should you do?

A. Adopt an “agile” development process. Reduce the time between releases as much as possible. Automate the build process from a source repository, which includes versioning and self-testing. Use Cloud Monitoring, Cloud Logging, and Cloud Alerting to ensure visibility. Use a canary deployment to detect issues that could cause rollback.

B. Adopt a “waterfall” development process. Maintain the current release schedule. Ensure that documentation explains how all the features interact. Ensure that the entire application is tested in a staging environment before the release. Ensure that the process to roll back the release is documented. Use Cloud Monitoring, Cloud Logging, and Cloud Alerting to ensure visibility.

C. Adopt an “agile” development process. Maintain the current release schedule. Automate build processes from a source repository. Automate testing after the build process. Use Cloud Monitoring, Cloud Logging, and Cloud Alerting to ensure visibility. Deploy the previous version if problems are detected and you need to roll back.

D. Adopt a “waterfall” development process. Maintain the current release schedule. Ensure that documentation explains how all the features interact. Automate testing of the application. Ensure that the process to roll back the release is well documented. Use Cloud Monitoring, Cloud Logging, and Cloud Alerting to ensure visibility.

Q80. You are an architect designing the disaster recover (DR) strategy for a solar company whose solution runs on Google Cloud Platform. Given the mission critical nature of the application, the customer wants low Recovery Point Objective (RPO) and Recovery Time Objective (RTO). The primary deployment site runs in africa-south1-b with the DR site in europe-west1. Given the the short window of RPO and RTO, you are concerned about the time it takes in exporting and importing the snapshots of persistent disks between the primary and DR regions. On

further analysis, you learn about a feature of GCE that addresses the concern of the cross-region copy of snapshots. Which feature is that?

A. Snapshots are global resources in GCE. They are available across all the regions by default.
B. Google Cloud Platform offers Disaster Recovery as a Service that doesn't require copying the snapshots across regions.
C. Persistent disks are global resources in GCE. They are available across all the regions by default. The same disk can be mounted on the instances running in primary and DR sites.
D. Google Developer Console has a scheduler that can automatically copy snapshots between regions.

Q81. A global news outlet gets 30% of their advertising revenue from online channels. Their news website serves customers in multiple geographic regions. They use Cloud CDN to provide a better experience for their users.
The head of the media division thinks that there is an opportunity to reduce the cost of operations of the website. Their CDN costs are a major concern, and they are looking for ways to optimize the cost without impacting the user experience.
They have asked for your recommendation on how to optimize the CDN costs. (Choose two)

A. Enable caching for all the common images and non-personalized content
B. Enable caching for all the content on the news site
C. Optimize images by resizing and formatting them appropriately
D. Use high resolution images for all the content and use multi region cloud storage to store them

Q82. Which data storage service provides data warehouse services for storing data but also offers an interactive SQL interface for querying the data?
A. Datalab
B. Dataproc

C. BigQuery
D. Cloud SQL

Q83. What is one benefit of applying firewall rules by tag rather than by address?
A. When a VM is created with a matching tag, the firewall rules apply irrespective of the IP address it is assigned.
B. Tags help organizations track firewall billing.
C. Tags on firewall rules control which ephemeral IP addresses VMs will receive.
D.Tags in network traffic help with network sniffing.

Q84. A web hosting company has built a three-tier application hosting critical workloads that they plan to host on the GCP. Currently, the database is designed as a part of the application server. Your manager has asked you to design the infrastructure in a way that follows the Hot Disaster Recovery Pattern.
How would you design the architecture?

A. Deploy one compute engine in one zone, failover to on-premises server.
B. Deploy two compute engines in two zones, each located in different regions.
C. Deploy two compute engines in two zones each located in the same region.
D. Deploy two compute engines in two zones, each located in the same region but in different projects.

Q85. Your organization has successfully migrated to Google cloud and wants to analyze their data stream to optimize operations. They do not have any existing code for this analysis, so they are exploring all their options. These options include a mix of batch and stream processing, as they are running some hourly jobs and live- processing some data as it

comes in.
Which technology should they use for this?

A. Google Cloud Dataproc
B. Google Cloud Dataflow
C. Google Container Engine with Bigtable
D. Google Compute Engine with Google BigQuery

Q86. You have installed a patch update on a group of Google Shielded VMs that have hosted your application for the last three weeks. Before rebooting the VMs, what step should you take to ensure that the VM security features operate correctly?

A. Update the integrity policy baseline
B. Disable the secure boot feature
C. Back up the secure memory boot logs
D. Rotate your encryption and signing keys

Q87. Developers of a microservice are receiving reports that their recently updated Google App Engine application is taking approximately 30 seconds to load for some of their users.
This behavior was not reported before the update.
What strategy should they take?

A. Work with your ISP to diagnose the problem
B. Open a support ticket to ask for network capture and flow data to diagnose the problem, then roll back your application
C. Roll back to an earlier known good release initially, then use cloud operations suite to diagnose the problem in a development/test/staging environment
D. Roll back to an earlier known good release, then push the release again at a quieter period to investigate. Then use Cloud operations suite to diagnose the problem

Q88. A utilities supplier has partially migrated to the cloud and still maintains many of its servers in an on-premises data center.
The supplier detects increased throughput latency when cloud applications communicate with corporate on-premises servers. While investigating the issue, you confirmed that load balancers are in place and configured correctly.
How might you adjust the configuration of this hybrid environment to minimize latency and improve application performance?

A. Enable window-size scaling from the command line to increase bandwidth.
B. Enable Round-Trip Time optimization from the command line to increase bandwidth.
C. Do nothing. This latency is necessary and acceptable because it prevents a backlog of packets consuming the resources of the receiving server.
D. After calling the [RTT.insert] method, check the RTT status by calling [RTT.get] with the job ID and location, and check the [status.state] value to learn the job status.

Q89. As an architect in a real estate company, you are troubleshooting a network issue. Network traffic between one Compute Engine instance and another instance is being
dropped. What is the likely cause?

A. A firewall rule that the instances depended on was deleted
B. The instances are on a default network, and nobody created additional firewall rules.
C. Network bandwidth is low
D. Your TCP/IP settings are improper.

Q90. The security team in your organization has discovered a rogue network that was created in your project. This network has an instance with an SSH port open to the world. They're working on identifying

where this network originated from. Where should they look?

A. Look in the Cloud Monitoring console in your project
B. Connect to the VM and look at the system logs to identify who logged into the environment.
C. Look through the logs in the console and specify GCE Network as the logging section.
D. Look at the Cloud Audit logs and determine which user accessed this environment the most. Then interrogate that user.

Q91. You are working in a hospital that has its business-critical applications hosted in GCP. As your users' engagement is increasing, cloud infrastructure requirements are also increasing. You are planning to configure additional infrastructure for another project using reserved instances to get a discount. You would need to set up networks, and all other resources required to host a basic web application. Due to a lack of employees, you need an easy way to continuously re-deploy certain infrastructure. What's the best way to do this?

A. Use Forseti Security to get configuration details from existing projects.
B. Make use of Cloud Deployment Manager.
C. Create resources one by one.
D. Use Asset History from Resource Manager.

Q92. You have an application running on a Compute Engine instance in a single VPC spread across two regions. This application needs to communicate over VPN to an on-premises network. How would you deploy the VPN?
A. Expose the VPC on-premises using VPC sharing and proper access controls.
B. Create a global VPN gateway and create a VPN tunnel for each region to your on-premises environment
C. Use VPC Network Peering to peer the VPC to your on-premises network

D. Deploy a Cloud VPN gateway in each region and ensure there is at least one VPN tunnel to the on-premises peering gateway

Q93. A weather site has applications running on VMs serving web traffic and you noticed that after you configured the VMs as a backend to an HTTP(S) load balancer, the VMs are being terminated and relaunched. What is most likely the best way to deal with the issue?

A. Ensure that a firewall rule exists to enable the health check on the load balancer to reach the instances in the instance group.
B. Assign a public IP to the instances and configure your firewall rules appropriately
C. Use network tags on each instance to route traffic from the instances to the load balancer.
D. Allow the source traffic on the instances to reach the load balancer by using firewall rules.

Q94. You want to calculate the uptime of a service and receive alerts if the uptime value falls below a certain threshold. Which tool will help you with this requirement?
A. Cloud Monitoring
B.Profiler
C. Error Reporting
D.Logs Explorer

Q95. Users have reported that an application you developed occasionally returns garbage data instead of the intended results, but you have been unable to reproduce this problem in your test environment. Which tool might be of the best help?
A. Error Reporting
B. Trace
C.Logs Explorer

D. Profiler

Q96. You're an architect designing an application that is going to be run for half a day only. You need to process batch jobs and would prefer to leverage a serverless product that scales to zero so that you can save money. What service should you use?

A. App Engine Flex
B. Cloud Functions
C. Kubernetes
D. Compute Engine PVMs

Q97. Your team is running Apache Spark and Hadoop jobs in their on-premises environment and would like to migrate to the cloud with the least amount of change to their tooling. What should they use?

A. Cloud Dataflow
B. Compute Engine with a Dataflow Connector
C. Kubernetes Engine with a Dataflow Connector
D. Cloud Dataproc

Q98. You're planning on migrating 8 petabytes of data to GCP. This data requires 24/7 availability, and your data analyst team is familiar with SQL. What tool should you use to surface this data to your analyst team for analytical purposes?

A. Cloud Datastore
B. Cloud SQL
C. Cloud Spanner
D. BigQuery

Q99. You are the cloud architect in an organization that has multiple projects used for development and testing. Each project has a budget, and each developer has a budget.
A personal budget overrun can cause a project budget overrun. Several developers are creating resources for testing as part of their CI/CD pipeline but are not deleting these resources after their tests are complete.
If the compute resource fails during testing, the test can be run again. You want to reduce costs and notify the developer when a personal budget overrun causes a project budget overrun. What should you do?

A. Configure billing export to BigQuery. Create a Google Cloud budget for each project. Create a group for the developers in each project and add them to the appropriate group. Create a notification channel for each group. Configure a billing alert to notify the group when their budget is exceeded. Modify the build scripts/pipeline to label all resources with the label "creator" set to the developer's email address. Use spot (preemptible) instances wherever possible.
B. Configure billing export to BigQuery. Create a Google Cloud budget for each project. Create a Pub/Sub topic for developer-budget-notifications. Create a Cloud Function to notify the developer based on the labels. Modify the build scripts/pipeline to label all resources with the label "creator" set to the developer's email address. Use spot (preemptible) instances wherever possible.
C. Configure billing export to BigQuery. Create a Google Cloud budget for each project. Create a Pub/Sub topic for developer-budget-notifications. Create a Cloud Function to notify the developer based on the labels. Modify the build scripts/pipeline to label all resources with the label "creator" set to the developer's email address. Use spot (preemptible) instances wherever possible. Use Cloud Scheduler to delete resources older than 24 hours in each project.
D. Configure billing export to BigQuery. Create a Google Cloud budget for each project. Configure a billing alert to notify billing admins and users when their budget is exceeded. Modify the build scripts/pipeline to label all resources with the label "creator" set to the developer's email address. Use spot (preemptible) instances wherever possible.

Q100. Cymbal Direct uses a proprietary service to manage on-call rotation and alerting. The on-call rotation service has an API for integration. Cymbal Direct wants to monitor its environment for service availability and ensure that the correct person is notified. What should you do?

A. Ensure that VPC firewall rules allow access from the IP addresses used by Google Cloud's uptime-check servers. Add the URL for the on-call rotation API as a monitoring notification channel in Google Cloud's operations suite. Create an uptime check for the appropriate resource's internal IP address, with an alerting policy set to use the API.
B. Ensure that VPC firewall rules allow access from the IP addresses used by Google Cloud's uptime-check servers. Create a Pub/Sub topic for alerting as a monitoring notification channel in Google Cloud's operations suite. Create an uptime check for the appropriate resource's internal IP address, with an alerting policy set to use the Pub/Sub topic. Create a Cloud Function that subscribes to the Pub/Sub topic to send the alert to the on-call API.
C. Ensure that VPC firewall rules allow access from the IP addresses used by Google Cloud's uptime-check servers. Create a Pub/Sub topic for alerting as a monitoring notification channel in Google Cloud's operations suite. Create an uptime check for the appropriate resource's external IP address, with an alerting policy set to use the Pub/Sub topic. Create a Cloud Function that subscribes to the Pub/Sub topic to send the alert to the on-call API.
D. Ensure that VPC firewall rules allow access from the on-call API. Create a Cloud Function to send the alert to the on-call API. Add Cloud Functions as a monitoring notification channel in Google Cloud's operations suite. Create an uptime check for the appropriate resource's external IP address, with an alerting policy set to use the Cloud Function.

Q101. You are deploying an application that needs to integrate with an on-premises database, but you don't want to expose your on-premises

database to the Internet. The performance requirements are not important. What's the quickest, most secure, and most cost-effective approach?

A. Use firewall rules to limit access to the on-premises database.
B. Use Cloud VPN to limit access to the on-premises database.
C. Use Partner Interconnect to limit access to the on-premises database.
D. Use Dedicated Interconnect to limit access to the on-premises database

Q102. You've been asked to lead the compliance aspect of your application design. To meet PCI-DSS requirements, what is the best approach to take?

A. Google already meets PCI-DSS by default, so most best practices are already implemented.
B. Turn on the compliance baseline API and determine whether there are any outstanding compliance issues for your application.
C. Gather the PCI-DSS requirements and then begin designing the application around the requirements your application needs.
D. Hire a contractor who has the PCI-DSS certification under his belt

Q103. You want to migrate your entire VM infrastructure to VMware Engine on Google Cloud. What sort of migration is this?
A. Replatforming
B. Rehosting
C. Refactoring
D. Rephasing

Q104. You need to transfer 40TB of data to Google Cloud in the most secure and operationally effective way. You have a 1 Gbps uplink to GCP. What should you do?

A. Use a transfer appliance.
B. Use the storage transfer service.
C. Directly upload the data to GCS
D. Provision a better uplink.

Q105. You need to reduce the number of unplanned rollbacks of erroneous production deployments in your company's web hosting platform. Improvement to the QA/
Test processes accomplished an 80% reduction.
Which additional two approaches can you take to further reduce the rollbacks? (Choose two.)

A. Introduce a green-blue deployment model
B. Replace the QA environment with canary releases
C. Fragment the monolithic platform into microservices
D. Reduce the platform's dependency on relational database systems
E. Replace the platform's relational database systems with a NoSQL database

1. **C**- Bigtable is ideal for IoT, gives consistent sub-10ms latency, and can be used at a petabyte scale.
2. **A-** To enable interactive access to an instance's serial console to debug boot and networking issues, you need to edit the metadata of the instance.
3. **A-** Cloud Functions can be directly triggered by changes (create, update, delete) to documents in Firestore, making it an ideal choice for this scenario. Cloud Run, on the other hand, is primarily designed for HTTP workloads and would require additional code to poll Firestore for changes.
4. **D-** A virtual machine is an object that describes what the emulated server should look like, while an instance is a running copy of the VM. A virtual machine describes the configuration of the emulated server, while an instance represents a running copy of that configuration.
5. **A**- Carrier Peering allows using a Google partner to connect to Google public infrastructure, such as Workspace or Gmail.
6. **A**- Configuring a Google Cloud Armor rule to prevent that IP address from accessing the HTTP-backend on the load balancer will prevent access.
7. **A**- Using separate service accounts for each component allows you to grant only the access needed to each service account with either a predefined or custom role.
8. **B**- BigQuery is a data warehousing service that allows the storage of huge data sets while making them immediately processable without having to extract or run the processing in a separate service.
9. **A- https://cloud.google.com/anthos/docs/concepts/overview**
10. **D-** Cloud Run is a managed microservices platform that allows us to just build our application into a container, then let Google worry about handling all the infrastructure management from there.
11. C- https://cloud.google.com/identity/docs/how-to/update-group-to-security-group
12. B- Integration with Google Cloud's operations suite streamlines and unifies these traditionally independent services, making it much easier to establish procedures around them and to use them in continuous ways.
13. C- Cloud Trace provides latency sampling and reporting for Google App Engine, Google HTTP(S) load balancers, and applications instrumented with the Cloud Trace SDKs
14. B- Services should be stateless, and a service like Memorystore for Redis provides a fast caching service to store state. They enable services to be stateless and support scale and high availability.
15. C- They have a standard Content-Type that can be set on the response header and are text-based. It is usual to use JSON, but both XML and JSON are valid

16. A- Cloud Build triggers have been designed specifically to trigger a build automatically when changes are made to source code.
17. C- A key feature of Spanner is scale for relational data with strong consistency, and it is globally distributed to provide low latency.
18. **C**-
19. **D**- Memorystore provides the best fit when considering data model, performance, scale, cost, and availability.
20. **D**- The purpose of load testing is not to break the systems, but to test the maximum number of users it could potentially support given the constraints of system resources. https://cloud.google.com/architecture/distributed-load-testing-using-gke
21. **C & D**-Signed urls are temporary urls that will prevent users from sharing the static url to access the content. CDNs help in distributing the content globally without replicating them in multiple cloud storage locations
22. **A, B & D**- https://cloud.google.com/blog/products/databases/introducing-cross-region-replica-for-cloud-sql
23. C- traffic is HTTP(S), the load balancer should be external and global, and CDN enabled will help performance and cost.
24. C- Cloud Spanner is a managed and scalable relational database from GCP. You can change the compute capacity of a Cloud Spanner instance but it doesn't scale automatically based on the load. The Autoscaler tool for Cloud Spanner can be used to automatically increase or reduce the number of nodes or processing units in one or more Spanner instances based on how their capacity is being used.
25. **D**- VPN offers a secure extremely reliable connection and is more cost-effective than Cloud Interconnect.
26. **D**- The best strategy is to use the canary deployment for A/B testing and then change the traffic split ratio based on the test result. The new feature should only be released to a small percentage of users for testing to avoid negative impact if the feature doesn't work well as designed.
27. **A**- Compute Engine offers full control over virtual machines including operating systems. No repackaging would be required.
28. **C-** Cloud Functions are for single purpose functions like image analysis. Cloud Functions also can be triggered by Cloud Storage events, so they provide seamless integration. The payment model based on number of requests, processing time of request (measured in 100ms units), and then other resources consumed is the most suitable of all options offered above. There is a free tier

too. Cloud Functions also provide automatic scaling, high availability, and fault tolerance.

29. **D**- https://cloud.google.com/architecture/fluentd-bigquery
30. D- GKE will help with the resource efficiency and hence cost, automate many aspects of the container management, and provide the best solution for the scenario.
31. B- Google's Virtual Desktop cloud enables access to corporate resources that is secure and scalable and for these reasons is the most appropriate solution in this scenario. Virtual Desktop includes User authentication and authorization with Google Workspace, IAP, or Active Directory.
32. D- circuit breaker will attempt to prevent an operation that is likely to fail and therefore will protect the resource that is in partial outage and hopefully prevent cascading failure.
33. C- Cloud Spanner meets all the requirements. It is a global relational database with high availability. Multi-regional instances have a monthly uptime of >=99.999%
34. A-
35. B- Cloud Storage always encrypts data on the server side before it is written to disk. For server side encryption there are options of customer supplied or customer managed encryption keys, but these are only usually used for compliance reasons and are not necessary.
36. **A, B & C**- Cloud VPN is the best way to connect the on prem system to the GCP as the data needs to be accessed only a few times. Creating a temporary VM in a separate VPC will make it more secure and avoid exposing the on-prem systems directly to the main VPC.The best way to achieve the required solution is to use the temporary VM to update the private keys whenever required and delete them after usage, so that they are not stored on the cloud.
37. **A-** A rolling update will update instances incrementally until all have been updated. This strategy can be controlled, for example, by using a managed instance group with Compute Engine or with GKE.
38. **A**- These are the most cost-effective solution as required in the question(up to 80% lower than equivalent non-preemptible machines) and the fact that some downtime is acceptable means that the requirement can be met with preemptible machines.
39. C- Availability is the percentage of time a system is running and able to process requests. Monitoring this metric will enable the derivation of the SLO metric.
40. B- Canary deployment aims to eliminate/reduce risks by applying the update to a small subset of users to test out a new feature and best fits the requirements of the question.

41. C-If the organization uses a third-party identity provider, the organizations' user directory should be synchronized with Cloud Identity to let users access Google Cloud with their corporate credentials. https://cloud.google.com/architecture/framework#groups-and-service-accounts
42. A- the backend setting for a high availability global load balancer to direct HTTP traffic across multiple zones will be as follows: The Backend service is web-app-backend. The Endpoint protocol is HTTP. The Health check is web-app-load-balancer-check. The Instance group is load-balancing-web-app-group
43. C
44. D
45. B- https://cloud.google.com/compute/docs/networking/tcp-optimization-for-network-performance-in-gcp-and-hybrid?hl=en
46. A- Direct Peering is the best choice. It is essentially free. Custom routing is not an option, and the connection is supported over the public internet rather than a private connection.
47. C- When converting physical server CPUs to Google Compute Engine vCPUs, it is important to remember that vCPU on a Compute Engine instance is implemented as a single hyper-thread on an Intel Xeon Processor. Since each Xeon Processor has two hyper-threads, that means you need to multiply the number of cores by two to get the number of threads.
48. C- Terraform lets you automate and manage resources in multiple clouds.
49. A
50. **B**- The security tab allows you to provide access to project resources to other users. Moreover, you can set up approval notifications via email or Cloud Pub/Sub.
51. **B-** Each step is dependent on the previous step. These are in the right order.
52. **D**- You can use network tags with firewall rules to automatically associate instances when they are created. Secure boot and vTPM protect the OS from being compromised.
53. **A**- Loose coupling has several benefits, including maintainability, versioning, and reduced complexity. Clients not knowing the backend systems means that these systems can be more easily replaced or modified, and HTTP(S) is the most common protocol used for external REST APIs.
54. D- Enabling VPC Service Controls lets you define a network perimeter. VPC Flow Logs lets you log network-level communication to Compute Engine instances.
55. B- Terraform lets you manage how you deploy and manage a variety of services in Google Cloud, such as Compute Engine. You can also use Cloud Deployment Manager for this purpose.
56. D- https://cloud.google.com/compute/docs/autoscaler

57. D- his will allow Kubernetes to scale the number of pods automatically, based on a condition like CPU load or requests per second.
58. C- This takes a lazy deletion approach and allows support or administrators to restore data later if necessary.
59. B-roles/bigquery.dataViewer is a sufficient role to grant viewer access to each country partner, so this choice is correct.
60. B- Cloud Billing export to BigQuery enables you to export detailed Google Cloud billing data (such as usage, cost estimates, and pricing data) automatically throughout the day to a BigQuery dataset that you specify.
61. B- You could use individual accounts to give out access instead of a group, and by doing so you make access more manageable. Identity-Aware Proxy is a great tool for exactly this kind of issue.
62. C- Because the environment has evolved, update the IAM resource hierarchy to reflect the changes. Use projects to group resources that share the same trust boundary.
63. B- Tags lets you reference the instances all at once, such as with a firewall rule. They can be used to manage the firewall rules where the instances are dynamically managed by a load balancer. https://cloud.google.com/resource-manager/docs/creating-managing-labels
64. C- The reports relating to compliance vulnerabilities are on the Compliance tab. To use the Security Health Analytics that scan for common compliance vulnerabilities, you must use the Premium tier.
65. D- Using Cloud NAT will prevent inbound access from the outside world but will allow connecting to social media APIs outside of the VPC. Using IAP or a bastion host allows for management by SSH, but without the complexity of using VPNs for user access.
66. C- Durability ensures that your data is protected and available. Snapshots are a viable way of backing up your data in Compute Engine.
67. D- Configuring a build trigger automates the CI/CD process based on when the software is posted to a repository.
68. C- Durability ensures that your data is protected and available. Snapshots are a viable way of backing up your data in Compute Engine.
69. B- Cloud Build triggers automate the build process when new files are placed into the name and branch of the repository that you specify.
70. C- A circuit breaker limits requests based on a threshold that you specify.
71. **B**- A/B testing is a pattern that lets you evaluate new proposed functionality.

72. **A & D** AI platform supports continuous model evaluation and Vertex AI supports data skew or drift detection. The model evaluation feature will be in the future release of Vertex AI.
73. **B**- While we can quickly end up with dozens or more network end points in between all of our microservices, with Anthos Service Mesh, we can again use a simple YAML configuration file to define how all our containers interact with each other, and with external network connections.
74. **A & B-** The company needs to store the variables in a BigTable to create insights and display metrics on the app to minimize latency. Data can be stored in cloud storage and the required variables can be sent into BigTable via. Dataflow connector
75. A & C- https://cloud.google.com/architecture/de-identification-re-identification-pii-using-cloud-dlp
76. A-To create a highly available and fault tolerant architecture, the servers must be deployed in 2 zones. Two server instances should be created in each zone / subnet and should be configured to allow traffic between them. https://cloud.google.com/architecture/deploy-an-active-directory-forest-on-compute-engine
77. **C**- An SLO of no more than 3% downtime over the course of a day would mean that a downtime of more than 43 minutes would exceed it.
78. **B**- Retention policies can provide immutable storage on Cloud Storage. In conjunction with Detailed audit logging mode, which logs Cloud Storage request and response details, Bucket Lock can help with regulatory and compliance requirements, such as those associated with FINRA & SEC
79. **A-** A modern CI/CD pipeline lets you release smaller changes more frequently and includes integrated testing. Using a canary deployment can let you detect issues before you deploy your new version at scale.
80. **A-** Snapshots are global resources which means they are available to all regions globally. This avoids the need to manually export and import snapshots between regions. A snapshot that was taken from a persistent disk attached to an instance in africa-south1-b will be instantly available in europe-west.
81. **A & C**-Multi region cloud storage is not an option for this use-case as it might impact the user experience. Caching should be enabled for common content and not for personalized content - there is no advantage in caching the personalized content as it is likely to be shown to a single user.
82. **C**- BigQuery is a data warehousing service that allows the storage of huge data sets while making them immediately processable without having to extract or run the processing in a separate service.

83. **A**- When a VM is created the ephemeral external IP address is assigned from a pool. There is no way to predict which address will be assigned, so there is no way to write a rule that will match that VM's IP address before it is assigned. Tags allow a symbolic assignment that does not depend on order in the IP addresses. It makes for simpler, more general, and easier to maintain, firewall rules.
84. **B**- deploy two compute engines in two zones, each located in different regions. This is correct because Hot DR Pattern has low RTO/RPO which means Fastest recovery. If you create a compute engine in 2 zones, if the Compute Engine goes down in 1 zone or 1 region, the other zone or region is still there. It is the most expensive approach.
85. **B**- Cloud Dataflow is a fully managed service for transforming and enriching data in stream (real time) and batch (historical) modes with equal reliability and expressiveness -- no more complex workarounds or compromises needed.
86. A- Updating the integrity policy baseline sets the baseline to the measurements captured from the most recent boot sequence. If it is not expected, you should stop that VM instance and investigate the reason for the failure.
87. C- Google Cloud's operations suite Logging allows you to store, search, analyze, monitor, and alert on log data and events from Google Cloud Platform
88. **A-** https://cloud.google.com/compute/docs/networking/tcp-optimization-for-network-performance-in-gcp-and-hybrid?hl=en
89. A- firewall rule was accidentally deleted or intentionally deleted because someone didn't understand the dependencies of the rule. Check to see if the instances are permitted to communicate with each other.

90. C- Look through the logs in the console and search for a Create Insert; it'll display the JSON code string that contains the e-mail used to create the network. Your security team can use this as part of their investigation to determine the origin and root cause.
91. B-Infrastructure as a Code helps you deploy the same architecture with the same configuration within a few minutes. You won't even have to review using Cloud Deployment Manager's IAC feature. You have to create an infrastructure template for the first time, and later on, you can use it directly.
92. D- deploy a VPN gateway in each region and deploy at least one tunnel to your on-premises environment, so that instances in other regions can use the tunnel for egress traffic and you can meet your high-availability requirements.
93. A- You are getting the appropriate web response from each instance, but the instances keep getting terminated and relaunched. The network pattern between

the load balancer health checks and the instances may not be properly configured.

94. A
95. A
96. **B**- You would want to leverage Cloud Functions in this instance because it is serverless, it can be leveraged to respond to events, and it scales to zero. Preemptible VMs are commonly associated with batch jobs, but you don't get to decide when a PVM is going to be run or not, and PVMs don't scale to zero.
97. **A**- Dataflow is a serverless solution that can be leveraged for both batch and stream processing. Dataproc is not fully serverless.
98. **D**- Large-scale migration, requirement to use SQL, and an analytical use case
99. **B**- You can have billing notifications sent to a Pub/Sub topic that triggers a Cloud Function. The function can then notify the appropriate developer.
100. **C**- Using Pub/Sub as a notification channel gives you flexibility to adapt how notifications are sent.
101. **B**- The fastest, most secure, and most cost-effective approach, since the performance requirements are not important, would be to leverage Cloud VPN to set up a private tunnel between the on-premises database and the application.
102. **A**- One advantage of GCP is that it's already compliant with many industry regulations.
103. **B**- By rehosting your platform on GCP, you can retain the use of your existing platform but move it over to Google Cloud. VMware Engine on GCP enables you to migrate your existing VM-based architecture on-premises to GCP
104. **A-** The key thing here is that you're transferring a significant amount of data to GCP and using a 1 Gbps uplink may not be the fastest way to do this. It's recommended that you use a transfer appliance when terabytes of data need to go to GCP.
105. **A & C**